

What Mary Really Learned: A Structural Interpretation of the Knowledge Argument Through the Theory of Axiomatic Necessity

Abstract

Frank Jackson's famous Knowledge Argument, commonly known as Mary's Room, remains one of the most influential challenges to physicalism. The central question is deceptively simple: if Mary knows every physical fact about color while living in a monochromatic environment, does she learn something genuinely new when she first experiences the color red?

This paper proposes a structural reinterpretation of the argument using the Theory of Axiomatic Necessity (TNA). Rather than viewing Mary's discovery as the acquisition of new factual information or merely a new cognitive ability, we argue that it represents a transition between two structurally distinct domains. Physical knowledge belongs to the operational domain (N_0), whereas conscious experience requires access to an admissibility domain (N_1) that cannot be generated solely through operational descriptions.

Within this framework, Mary's first perception of red is interpreted as a structural transition governed by the TNA operator Ψ , which maps operationally represented states into admissible conscious experience. Consequently, the Knowledge Argument becomes another instance of the TNA Closure Theorem: no operational system can derive its complete admissibility conditions from within itself.

Keywords: Theory of Axiomatic Necessity, consciousness, Mary's Room, Knowledge Argument, physicalism, qualia, admissibility, Failure of Local Closure.

1. Introduction

Among all thought experiments in the philosophy of mind, few have generated as much discussion as Frank Jackson's *Mary's Room*.

Mary is a neuroscientist. She knows every physical fact about color vision. She understands optics, neurophysiology, electromagnetic radiation, and visual cortex dynamics—every measurable physical property associated with color perception. Yet she has lived her entire life inside a perfectly monochromatic environment. She has never experienced color.

The question is simple: When Mary leaves the room and sees a red rose for the first time, does she learn something new?

Jackson originally answered yes. If she does, then complete physical knowledge appears insufficient to exhaust conscious experience. This paper argues that the significance of Mary's discovery is neither epistemic nor merely phenomenological. It is structural.

2. The Knowledge Argument

The traditional formulation distinguishes two positions:

- **Physicalism:** Maintains that complete physical knowledge completely describes reality.
- **The Knowledge Argument:** Suggests otherwise. Mary already possesses complete physical knowledge. Nevertheless, when she sees red for the first time, something changes.

The central dispute has traditionally been framed as the acquisition of a:

- New fact
- New ability
- New representation
- New mode of access

This paper proposes another possibility: Mary undergoes a transition between distinct structural domains.

3. Information Versus Admissibility

Mary already knows:

- The wavelength corresponding to red
- Retinal phototransduction

- Opponent-process theory
- Cortical color processing
- Every physical equation describing color perception

Nothing new is added to the physical description. No new scientific data appear. No additional neurons suddenly exist. Yet Mary's conscious state changes.

This distinction suggests that informational completeness and experiential admissibility are not identical concepts. Knowing every operational relation does not necessarily imply access to every admissible experience.

4. A TNA Reformulation

The Theory of Axiomatic Necessity distinguishes between:

- N_0 : Operational domain.
- N_1 : Admissibility domain.

Operational systems execute processes. Admissibility domains define the conditions under which those processes become meaningful or experientially accessible.

In Mary's case:

- N_0 contains: physics, neuroscience, computational models, and symbolic descriptions.
- N_1 contains: admissible conscious experience.

Mary's knowledge before leaving the room belongs entirely to N_0 . The experience of red belongs to N_1 .

5. The Structural Transition

Mary's experience should not be interpreted as receiving additional physical information. Instead, it is the first operational realization of a previously inaccessible admissible state.

We define Σ_{phys} as the space of physically describable cognitive states, and Σ_{qualia} as the space of admissible phenomenal states. The transition is represented by:

$$\Psi : \Sigma_{\text{phys}} \rightarrow \Sigma_{\text{qualia}}$$

where Ψ denotes the structural activation through which operationally represented information becomes conscious experience. The crucial point is that Ψ is not reducible to the operational description itself; otherwise, Mary would already possess the experience while still inside the room.

6. Why Physical Information Is Insufficient

Physicalism often assumes that sufficiently complete operational knowledge eventually produces consciousness. Mary challenges precisely this assumption. Her case demonstrates that complete operational representation does not automatically produce admissible experience.

Within TNA, this corresponds to:

$$N_0 \not\vdash N_1$$

Operational completeness does not entail admissibility. This is the **Failure of Local Closure**.

7. The Nature of Qualia

Qualia are often described as ineffable subjective properties. Within the present framework they may instead be understood structurally.

A quale is not an additional physical object, nor is it supernatural. It is the realization of an admissible experiential state. The redness of red is therefore not another wavelength; it is the admissibility of experiencing that wavelength. This distinction separates operational description from experiential realization.

8. Connections with Artificial Intelligence

Mary's Room has acquired renewed importance in the age of artificial intelligence. Large Language Models manipulate symbols with extraordinary sophistication. They can describe color, explain optics, discuss visual neuroscience, and even generate poetry about sunsets.

Yet an important question remains: Do they experience red?

Operational competence alone cannot answer this question. A system may perfectly manipulate symbolic representations while lacking the admissibility conditions required for conscious experience. This parallels Mary's condition before leaving the room. Complete operational knowledge does not necessarily imply conscious realization.

9. Relation to Gödel

The structural similarity with Gödel's incompleteness theorem is striking. Gödel demonstrated:

$$F \not\vdash \text{Con}(F)$$

TNA generalizes this relation. Mary's Room provides another manifestation of this asymmetry:

$$N_0 \not\vdash N_1$$

The operational domain cannot completely derive the admissibility conditions that make conscious experience possible. The similarity is structural rather than semantic.

10. Relation to the Software Gap

The present argument naturally complements the Software Gap hypothesis:

- **In human evolution:** Anatomically modern brains existed long before symbolic civilization emerged.
- **In Mary's case:** Complete physical knowledge exists before conscious realization occurs.

Both reveal the same asymmetry. Potential precedes realization; admissibility precedes operation. Civilization required culture; Mary required experience. Neither could be generated solely through operational completeness.

11. Discussion

The present interpretation differs from both classical dualism and reductive physicalism. It does not propose an immaterial substance, nor does it deny neuroscience. Instead, it distinguishes explanatory layers.

Operational descriptions remain indispensable, yet they do not exhaust admissibility. The Knowledge Argument therefore becomes a structural argument rather than merely an epistemological one. What Mary lacks is not additional information; she lacks access to an admissible experiential domain. When she leaves the room, she does not discover another physical law—she undergoes a structural transition.

12. Conclusion

What did Mary really learn? Not another scientific fact, not another equation, and not another wavelength. She learned what red is like because, for the first time, she entered the admissibility domain in which red can be consciously experienced.

Within the Theory of Axiomatic Necessity, this transition is represented by the operator:

$$\Psi : \Sigma_{\text{phys}} \rightarrow \Sigma_{\text{qualia}}$$

and constitutes another instance of the TNA Closure Theorem. Under this interpretation, Mary's Room is no longer merely a challenge to physicalism. It becomes a structural demonstration that operational completeness cannot, by itself, generate the admissibility conditions required for conscious experience:

$$N_0 \not\vdash N_1$$

The Knowledge Argument is therefore not an isolated paradox of philosophy of mind. It is another manifestation of a universal structural asymmetry that appears in mathematics, language, artificial intelligence, cosmology, cultural evolution, and consciousness itself.

References

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Appendix: The Software Gap as TNA Instance

TNA CONCEPT	SOFTWARE GAP CORRESPONDENCE
N_1 (Admissibility)	Human brain: space of cognitive possibilities
N_0 (Operational)	Culture: realized symbolic trajectories
Σ	Unrealized cognitive potential
σ^*	Actualized symbolic civilization
Ψ	"Installation" of cultural operating system
Failure of Local Closure	$N_0 \not\subset N_1$: culture cannot generate the brain's potential from within culture
TNA Closure Theorem	No operational system derives its complete admissibility conditions